YASH LALA

yashlala.com ◇ github.com/yashlala ◇ linkedin.com/in/yashlala
(510)-400-5572 ◇ yash@yashlala.com ◇ Palo Alto, CA

OVERVIEW

I'm a fourth-year PhD student at Yale, broadly interested in systems for efficient data-center scale computing. My research projects are generally themed around tradeoffs between memory capacity/bandwidth requirements and application performance. I approach these problems from a few different angles.

My primary specialty is in low-level operating systems and network stack optimization (via RDMA-based far memory systems on the CPU side, and MoE network performance modelling on the GPU side). I have 5+ years of kernel programming expertise, including large-scale projects such as reimplementing the memory-management subsystem in my most recent publication (SOSP '25). I have demonstrated expertise in developing high-throughput, low-latency kernel datapaths — such as a Linux-based system capable of migrating 200 Gib/s of memory at <270 ns per page per core. I'm deeply familiar with CPU microarchitecture and its interactions with the OS, and excel in principled, data-driven, bottleneck analyses.

Recently, I've also been tackling these problems from a high-level algorithm design/theory perspective. I feel that expertise in both worlds is necessary for effective algorithm-system codesign. I'm currently developing compact data structures for efficient nearest-neighbor queries in high-dimensional vector spaces. This project aims to accelerate RAG applications by marrying theory and practice: we provide formally proven runtime bounds, while maintaining strong empirical performance over billions of compressed RAG embeddings.

EDUCATION

Yale Ph.D. in Computer Science — Ongoing
UCLA B.S. in Computer Science — GPA: 3.782

PUBLICATIONS

Scalable Far Memory: Balancing Faults and Evictions — August 2023 - August 2025
31st Symposium for Operating Systems Principles, (SOSP 2025)

- Adapted the Linux Kernel to create a far-memory system capable of offloading excess application memory to arbitrary storage backends, for cluster-wide memory utilization improvements. Unlike prior work, this system scales to highly threaded, memory-intensive applications with minimal (near-ideal) application slowdown. Results are applicable to tiered memory systems and all far-memory backends (eg. NVMe, zswap, and RDMA).
- Rearchitected the Linux memory management subsystems to handle high-throughput page faulting and eviction. System achieves >200 Gib/s page migration throughput with only two CPU cores.
- Extensive low-level performance work around the OS-CPU interface, extending well into microarchitectural details. Gained expertise in: x86 IPI overheads and queuing characteristics; IOTLB-related virtualization overheads; TLB and page-walker coherence on x86; RDMA NIC control path throughput bottlenecks due to driver bugs; hypervisor-mediated IPI delivery costs; and lockless programming within the kernel.
- Developed principles for scalable memory migration. Results generalize to a Library-OS based re-implementation.
- Co-first author.

GRU4RecBE: Session Based Recommendations with Features — 2022
AAAI-22 Student Track

- Developed session-based recommendation system in PyTorch which extends the GRU4REC architecture with rich item features extracted from the pre-trained BERT architecture. Non-attentive model outperforms state-of-the-art session-based models over the MovieLens benchmark datasets.

PROFESSIONAL EXPERIENCE

NOVA Lab, Yale CS Department
PhD Student — *Advisor: Anurag Khandelwal*

- Broadly researching systems for efficient data-center scale computing. Recent work focuses on far-memory systems.
- Developed techniques for accelerating OS memory migration, aimed towards facilitating high-throughput RDMA-based far memory runtimes. Developed extensive knowledge of the the Linux kernel's memory management, RDMA, page placement, IPI, and swap subsystems. My particular focus was on optimizing high throughput, low-latency runtimes, such as memory migration systems and RDMA SmartNIC network stacks. Published in **SOSP '25**.
- Developing compact data structures for efficient range queries in high-dimensional vector spaces. Project is intended to accelerate floating-point vector databases for Retrieval-Augmented Generation (RAG) applications. Paper under submission.

NVIDIA
Systems Software Intern *Supervisor: Misha Smelyanskiy*

- Developed performance models for Mixture-of-Experts (MoE) communication phases for AI/ML models on NVIDIA's GB200 cluster hardware. Estimated ROI for future communication optimizations, and localized network performance bottlenecks in distributed systems of GPUs.
- Examined MoE token to expert mapping distributions to quantify the workload changes required for optimal hardware utilization. Explored algorithm-network codesign opportunities.
- Profiled existing GPU MoE library implementations (such as NCCL, DeepEP, etc). Optimized GPU MoE libraries for NVIDIA hardware by better utilizing tiered Multi-Node NVLink and Infiniband datacenter networks.
- Gained experience with LLM workload network traffic patterns and GPU performance debugging (particularly in GPU-initiated RDMA network communication).

SOLAR Lab, UCLA CS Department
Student Researcher *Supervisor: Harry Xu*

- Developed Linux kernel mechanisms for transparent far memory, with the goal of merging my changes upstream. The patchset extends the cpuset controller to allow per-cgroup control of active swap devices. The associated refactoring has positive implications for swap throughput, and makes it easy to manage frontswap-based remote memory systems. Patchset available at github.com/yashlala/canvas-linux.
- Developed a patchset to improve the Linux kernel's physical page allocation tail latencies by refilling the percpu low-order free page lists asynchronously using RCU.

CSSI Program, UCLA CS Department
Tutor Undergrad (TA)

- Taught introductory data science to high school students. Led discussion sections, prepared assignments, graded papers, and advised students.

Veritas Technologies LLC
SDE Intern

- Worked on NetBackup Flex, a platform for large-scale data consolidation and backup. Implemented automatic node discovery and cluster assimilation over datacenter networks. Primarily worked with Ansible, Docker, and various glue languages.

Pringle Lab, Stanford Genetics Department June 2017 - August 2017
Undergraduate Research Intern

- Tested algal species for selective binding to various lectin proteins in order to understand the chemical processes behind coral bleaching. Poster available at yashlala.com/pringle-poster.pdf.

PROJECTS

NDN Multicast

- Extended routing protocols for NDN (Named Data Network) under Lixia Zhang. Extended NLSR (a link-state routing algorithm for NDN) to allow for efficient multicast delivery of NDN Interest packets. Student paper available at yashlala.com/nlsr-poster.pdf.

MISCELLANEOUS

Enjoys teaching: led student seminar series, volunteered with ACM TeachLA and as a docent for the Computer History Museum. Certified as a operator for the punched-card based IBM 1401.